REQUIREMENT ANALYSIS PHASE

1. PHASE OVERVIEW

The Requirement Analysis Phase focuses on identifying and defining the requirements of the Auto Ticket Classification using Flow Designer project.

The school IT helpdesk receives multiple support requests from students and teachers related to network connectivity, hardware failures, account access problems, and system performance issues.

Currently, IT staff manually review each ticket, identify the issue type, select the appropriate Category and Subcategory, and inform the caller about ticket creation.

The proposed system is designed to automate these activities using ServiceNow Flow Designer.

2. PROBLEM IDENTIFICATION

The existing manual ticket classification process requires IT staff to review every support request and manually assign the appropriate Category and Subcategory.

This process can be:

Time-consuming.

Error-prone.

Difficult to manage when the number of support requests increases.

Dependent on manual classification by IT staff.

Therefore, an automated ticket classification system is required to improve the efficiency and consistency of IT support ticket processing.

3. PROJECT REQUIREMENTS

The system must satisfy the following requirements:

1. Automatically classify IT tickets based on the issue description.
2. Assign both Category and Subcategory without manual intervention.
3. Support dependent choice logic between Category and Subcategory.
4. Send an automated email notification to the caller upon ticket creation.
5. Store ticket information in a structured and standardized format.
6. Ensure easy maintenance and future scalability.

4. FUNCTIONAL REQUIREMENTS

The functional requirements of the system describe the operations that the application must perform.

4.1 Automatic Ticket Classification

The system should automatically identify the type of IT support issue from the ticket description.

4.2 Automatic Category Assignment

The system should automatically assign the appropriate Category based on the identified

issue.

4.3 Automatic Subcategory Assignment

The system should automatically assign the corresponding Subcategory.

4.4 Dependent Category and Subcategory

The Subcategory field should depend on the selected Category so that the appropriate choices can be maintained.

4.5 Automated Email Notification

After ticket creation, the system should send an automated email notification to the caller.

4.6 Structured Ticket Storage

The system should store ticket information in a structured and standardized format.

5. TICKET CLASSIFICATION REQUIREMENTS

The project requires classification of common school IT support issues.

The required classification mapping is:

Issue | Category | Subcategory

Wi-Fi / Network | Network | Wi-Fi

Projector | Hardware | Projector

Password / Login | Access | Forgot Password

Slow / Hanging Computer | Performance | Slow Computer

These classification requirements form the basis for the Flow Designer automation implemented in the later development phase.

6. DATA REQUIREMENTS

The system requires structured ticket information to perform classification and maintain ticket records.

The required information includes:

Ticket Number

Caller

Category

Subcategory

Short Description

Description

State

Assigned Group

Assigned To

These fields support the storage and management of IT support ticket information.

7. CATEGORY REQUIREMENTS

The Category field should support the following values:

Network

Hardware

Access

Performance

These categories represent the major types of IT support issues handled by the system.

8. SUBCATEGORY REQUIREMENTS

The Subcategory field should support the following values:

Wi-Fi

Projector

Forgot Password

Slow Computer

The Subcategory values should be associated with their corresponding Categories.

9. DEPENDENCY REQUIREMENT

The system must support a dependency relationship between Category and Subcategory.

The required dependency mapping is:

Subcategory | Category

Wi-Fi | Network

Projector | Hardware

Forgot Password | Access

Slow Computer | Performance

This requirement ensures that the available Subcategory options are related to the selected Category.

10. NOTIFICATION REQUIREMENT

The system should send an automated email notification to the caller after a ticket is created and classified.

The email notification should contain a confirmation message regarding the submitted support request.

This requirement ensures that the caller receives confirmation without requiring manual communication from the IT staff.

11. NON-FUNCTIONAL REQUIREMENTS

The project should also satisfy the following general requirements:

11.1 Maintainability

The solution should be easy to maintain and update when new ticket classifications are required.

11.2 Scalability

The system should support future expansion with additional categories, subcategories, and classification conditions.

11.3 Consistency

The system should apply the same classification logic to tickets containing the defined issue keywords.

11.4 No-Code Implementation

The automation should be implemented using ServiceNow Flow Designer, providing a no-code approach that is easy to maintain.

12. USER REQUIREMENTS

The main users involved in the system are:

IT Helpdesk Staff

IT staff require a system that automatically classifies incoming support tickets and reduces the need for manual classification.

Students and Teachers

Students and teachers require a simple way to submit IT support requests and receive confirmation after ticket creation.

13. TECHNOLOGY REQUIREMENT

The project requires:

Platform: ServiceNow

Automation Tool: Flow Designer

Application: Global

Ticket Management: Custom Incident Workflow table

Notification: Flow Designer Send Email action

These technologies support the implementation of the required ticket classification and notification functionality.

14. REQUIREMENT SUMMARY

Requirement | Description

Automatic Classification | Classify tickets based on issue description

Category Assignment | Automatically assign Category

Subcategory Assignment | Automatically assign Subcategory

Dependency | Maintain Category–Subcategory relationship

Email Notification | Notify caller automatically

Structured Storage | Store ticket information systematically

Maintainability | Support easy future changes

Scalability | Support additional classifications

15. PHASE 2 OUTCOME

At the end of the Requirement Analysis Phase, the major requirements of the project are clearly identified and documented.

The identified requirements include:

1. Automatic ticket classification.
2. Automatic Category assignment.
3. Automatic Subcategory assignment.
4. Category–Subcategory dependency.
5. Automated email notification.
6. Structured ticket information storage.
7. Maintainable automation.
8. Future scalability.

These requirements provide the foundation for the Project Design Phase, where the system structure and workflow will be designed.

16. CONCLUSION

The Requirement Analysis Phase defines the functional and non-functional requirements of the Auto Ticket Classification using Flow Designer project.

The analysis identifies the problems in the existing manual ticket classification process and defines the requirements for automating ticket classification, assigning Category and Subcategory, maintaining dependent choices, storing ticket information, and notifying callers.

The identified requirements provide a clear foundation for designing and developing the automated ServiceNow solution.